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EXPT NO. - 1

TITLE - SELECTION SORT

In [2]:

```
import random
import time
import matplotlib.pyplot as plt

# Selection Sort Function
def selection_sort(arr):
    n = len(arr)
    for i in range(n):
        min_idx = i
        for j in range(i + 1, n):
            if arr[j] < arr[min_idx]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
    return arr

# User Input
arr = list(map(int, input("Enter the list of elements separated by space: ").split()
print("Input array:", arr)

# Sorting
sorted_arr = selection_sort(arr.copy())
print("Sorted array:", sorted_arr)

# Time Measurement
start_time = time.time()
selection_sort(arr.copy())
end_time = time.time()
print("Time taken to sort:", end_time - start_time, "seconds")

print("=================================")

# For n > 5000
n_values = [5000, 6000, 7000, 8000]
time_values = []

for n in n_values:
    arr = [random.randint(1, 10000) for _ in range(n)]

    start_time = time.time()
    selection_sort(arr)
    end_time = time.time()

    time_taken = end_time - start_time
    print(f"Time taken to sort {n} elements: {time_taken} seconds")

    time_values.append(time_taken)

print("=================================")
```

```python
# Plotting Graph
plt.plot(n_values, time_values, color='green', marker='o')
plt.xlabel('Number of Elements (n)')
plt.ylabel('Time Taken (seconds)')
plt.title('Selection Sort Time Complexity Analysis')
plt.grid(True)
plt.show()
```

```
Input array: [5, 3, 8, 1, 2]
Sorted array: [1, 2, 3, 5, 8]
Time taken to sort: 0.00022292137145996094 seconds
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Time taken to sort 5000 elements: 3.2951269149780273 seconds
Time taken to sort 6000 elements: 3.5725464820861816 seconds
Time taken to sort 7000 elements: 4.44205904006958 seconds
Time taken to sort 8000 elements: 7.26636815071106 seconds
=================================
```